CSCE 611 Midterm Fall 2025

Name: _____ Score: ____/125

Part 1: RISC-V ISA

1. Consider the following snippet of RISC-V code:

```
loop: slli  x2,x2,1
      subi  x3,x3,1
      bne   x3,x0,loop
```

   a. (5 points) What does this code do?

   b. (5 points) How many instructions are executed when this code is executed with register x3 containing the value of 3?

   c. (5 points) Write an equivalent version of this code that performs the same function but after executing less instructions.

2. (10 points) Write a snippet of RISC-V assembly code (one or more instructions) that stores the value of 9.125 into register x2 as a (32,3)-format fixed-point number; that is, 32 total bits of which 3 bits are fractional (29 whole bits and 3 fractional bits).

Points: _____

Part 2: SystemVerilog

3. (20 points) Identify any problems (i.e. compile errors or bad practices) with the code below:

```
module bar (input logic [1:0] inp, output logic [2:0] outp);
  outp = inp;
endmodule

module foo (input logic [2:0] a,b, output logic [3:0] c);
 bar mybar(.inp(a),.outp(b));
endmodule
```

4. (5 points) To what value will the following SystemVerilog code set to signal b when a = 4'b1011?

```
module bit_manipulator (input [3:0] a, output [3:0] b);
  assign b = {4{a[0]}};
endmodule
```

5. (20 points) Write a SystemVerilog module named 'pulse18'. It accepts a clock signal named 'clk', an active-high asynchronous reset signal named 'rst', and a single-bit output named 'p'. When out of reset, the module will pulse the 'p' output (set to 1'b1 for one cycle) every 18 cycles.

Points: _____

6. (10 points) Convert the behavior of the following Verilog module into a truth table:

```
module test_module(input logic [2:0] a, output logic [1:0] y);
    always_comb begin
        y = 2'11;
        if (a[2]) y = 2'b00;
    end
endmodule
```

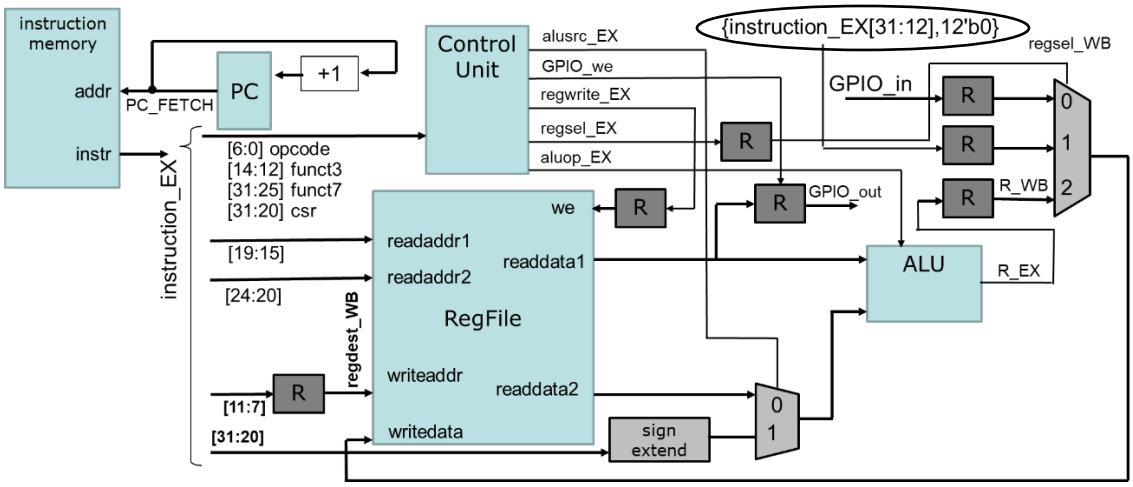
7. (5 points) How many clock cycles of delay will the code below impose between input "inval" and output "c"?

```
always_ff @(posedge clk) begin
  a <= inval;
  b <= a;
  c <= b;
  d <= c;
end
```

Points: _____

Part 3: Pipelined Microarchitecture

Consider the design of the 3-stage pipelined CPU design shown below:



a. (10 points) What is the purpose of the circled portion of the design?

b. (5 points) What is the purpose of the 'GPIO_we' signal?

c. (20 points) In the following table, mark the stage holding each of the instructions listed in the first four rows. Use the following abbreviations: fetch = "F", decode/execute = "E", write back = "W". Assume the instructions appear consecutively and the first instruction is fetched in cycle 0. Give the state of each of the control signals shown in the last five rows in each of the 6 cycles. Use "U" to denote an unknown value and "X" for don't care.

	cycle 0	cycle 1	cycle 2	cycle 3	cycle 4
`addi x2,x2,1`					
`mul x3,x3,x4`					
`add x3,x3,x5`					
regwrite_EX					
regwrite_WB					
GPIO_we					
regsel_EX					
regsel_WB					
alusrc_EX					

d. (5 points) Does this CPU support branch and jump instructions? Why or why not?

Points: _____